



# 23-DATASET-AND-CORPUS-SPEC.md — Every image has a licence and a name



**01 — Provenance or it does not train.** If you cannot name the source and the licence of an image, it cannot enter the corpus. This is not a legal formality; it is the only thing standing between AURA and an unfixable liability, because you cannot un-train a model.

## 1. The four corpora, and why they must never mix

| Corpus | Size target               | Purpose                                          | May a model ever see it?                  |
|--------|---------------------------|--------------------------------------------------|-------------------------------------------|
| TRAIN  | 60k–250k images           | Fitting model weights                            | Yes — this is the only one                |
| EVAL   | 8,000 images, 40 weddings | Ship gates in 18-MODEL-EVAL-GATES.md             | Never. Leakage test enforces this         |
| GOLDEN | 1,200 images, 3 weddings  | Determinism digests, CI                          | Never. Frozen forever                     |
| DEMO   | 300 images                | Screenshots, marketing, first-run sample project | Never. Needs the strictest licence of all |

**The split is by wedding, never by image.** Two photos of the same couple, taken ninety seconds apart, are near-duplicates; splitting them across TRAIN and EVAL inflates every metric and is the single most common way a model team lies to itself. Split at the level of `wedding_id`, and additionally hold out entire *photographers* so that the eval measures generalisation to a new style, not memorisation of one.

## 2. Permitted sources, in order of preference

1. **Contributed by partner photographers under a signed data agreement.** The gold standard. Aim for 25 photographers across at least 6 countries. Offer a free perpetual licence in exchange — this is the cheapest data acquisition available to you and it also produces your beta cohort.
2. **Public domain and CC0 archives** (Wikimedia Commons, museum releases, Unsplash under its licence terms — read them, they changed).
3. **Licensed stock**, purchased with a written model-and-property release for AI training. Most stock licences *exclude* AI training; assume excluded unless the contract says otherwise in writing.
4. **Synthetic and augmented data**, derived only from images already in category 1–3.
5. **Your own photography.** Unlimited, free, and the fastest way to fill a gap slice.

**Forbidden, without exception:** scraped Instagram, Pinterest, Flickr "all rights reserved", photographers' portfolio sites, wedding blogs, Google Images, any dataset of unclear origin, and any dataset whose licence permits "research use" only — AURA is a commercial product.

## 3. The data agreement, in plain terms

The partner photographer agreement must state, in language a photographer will actually read:

- What is contributed: RAW files, their edited JPEGs, and their cull decisions (the last is the most valuable and the least obvious).
- That they warrant they hold copyright and have model releases from the people depicted.
- That contributed images are used to train models, never republished, never shown to other users, and never used in marketing without a separate opt-in.
- That they may withdraw, and what withdrawal means honestly: images are removed from the corpus and excluded from all future training, but weights already trained cannot be surgically unlearned — removal takes effect at the next model version. Saying anything stronger is a lie.

- Retention period and deletion procedure.
- Compensation: perpetual licence tier, named credit if they want it.

## 4. The manifest — frozen contract

Every image in every corpus has a row. No row, no image; a CI job fails the build if a file exists on disk without a manifest entry.

```
-- corpus/manifest.sqlite -- FROZEN CONTRACT
CREATE TABLE corpus_image (
  image_uuid      TEXT PRIMARY KEY,
  sha256          TEXT NOT NULL UNIQUE,
  corpus          TEXT NOT NULL CHECK (corpus IN ('TRAIN', 'EVAL', 'GOLDEN', 'DEMO')),
  wedding_id      TEXT NOT NULL,
  photographer_id  TEXT NOT NULL,
  source_type      TEXT NOT NULL CHECK (source_type IN ('partner', 'public_domain', 'cc0', 'licensed_stock', 'synthetic', 'owned')),
  licence_ref      TEXT NOT NULL,    -- path to the signed agreement or licence text
  acquired_at      TEXT NOT NULL,
  model_release    INTEGER NOT NULL, -- 0/1, required for DEMO
  withdrawal_state TEXT NOT NULL DEFAULT 'active'
                                CHECK (withdrawal_state IN ('active', 'withdrawn')),
  -- slice axes, mirrored from 18-MODEL-EVAL-GATES.md
  skin_tone_monk    INTEGER,          -- 1-10, annotated not inferred
  lighting          TEXT,             -- daylight|golden|indoor_warm|candlelight|flash|mixed|night
  venue            TEXT,             -- outdoor|church|hall|beach|home|temple|mandap
  culture           TEXT,             -- self-declared by contributing photographer
```

```

camera_make      TEXT,
camera_model     TEXT,
motion          TEXT,          -- static|walking|dancin
g|confetti
group_size      INTEGER,
-- labels
label_set       TEXT,          -- JSON: which annotatio
ns exist
annotator_id    TEXT,
annotation_round INTEGER
);

CREATE TABLE corpus_split_lock (
wedding_id      TEXT PRIMARY KEY,
corpus          TEXT NOT NULL,
locked_at       TEXT NOT NULL,
locked_by       TEXT NOT NULL
);

```

`corpus_split_lock` is what makes leakage structurally impossible: once a wedding is assigned, it can never be reassigned. Moving a wedding from EVAL to TRAIN because "we need more data" is the exact failure this table exists to prevent, and it requires an ADR to override.

## 5. Coverage targets — the corpus is a product requirement

A model is only as fair as its worst-represented slice, and `18-MODEL-EVAL-GATES.md` enforces a 3 pp fairness spread. That gate is unachievable unless the corpus is built for it deliberately.

| Axis                   | Minimum requirement in EVAL                    | Why this number                                                                    |
|------------------------|------------------------------------------------|------------------------------------------------------------------------------------|
| Monk skin tone<br>1-10 | ≥ 400 faces per tone, ≥ 6<br>weddings per tone | Below this, per-slice metrics are<br>noise and the fairness gate is<br>meaningless |

| Axis           | Minimum requirement in EVAL                                                         | Why this number                                                                    |
|----------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Lighting       | ≥ 500 images per condition;<br>candlelight and night specifically ≥ 800             | The hardest conditions need the most evidence                                      |
| Culture        | ≥ 4 weddings each: South Asian, East Asian, African, Latin, Middle Eastern, Western | Red and gold saris, tea ceremonies and mandaps break Western-trained colour models |
| Camera         | ≥ 3 bodies each: Canon, Nikon, Sony, Fujifilm; ≥ 1 Leica, Panasonic, Pentax         | CFA patterns and colour science differ; RAW decode bugs are camera-specific        |
| Group size     | ≥ 200 images with > 30 faces; ≥ 20 with > 100                                       | Group shots are where face detection and eye-state detection collapse              |
| Disaster cases | 1 full wedding, deliberately bad: missed focus, blown highlights, wrong WB          | wedding-c-disaster in the golden set — the product must degrade gracefully         |

Publish the coverage table as a CI artefact. When a slice is under target, that is a **data bug** with an owner, tracked exactly like a code bug.

## 6. Annotation protocol

The cull labels are the heart of AURA, and they are subjective. Treat that honestly:

- **Three annotators per image** for cull decisions, drawn from working wedding photographers, not crowdworkers. A crowdworker cannot tell you whether a slightly soft photo of the first kiss is a keeper. It is.
- **Report inter-annotator agreement** (Krippendorff's alpha) per slice. If human agreement on "keep or cut" is 0.62, then a model at 0.62 is at human parity and demanding 0.85 is demanding that it be wrong in a consistent way.
- **Never resolve disagreement by majority alone.** Keep the distribution: an image where 2 of 3 said keep is a *soft* label, and training on soft labels is how you get calibrated confidence, which is what the autonomy bands in 08-AUTONOMY-MODEL.md depend on.
- **Golden-standard checks:** 5 % of each annotator's queue is pre-labelled by the lead. An annotator drifting below 80 % agreement on those is retrained or

removed.

- **Annotation rounds are versioned.** Re-labelling is a new `annotation_round`, never an overwrite, so a model regression can be traced to a label change.

## 7. Storage, cost and the offline reality

RAW files at 45 MP are ~55 MB each; 100,000 of them is 5.5 TB, which will not sit in a Git repository or on a laptop.

- **Tiered storage.** Full RAWs in cheap object storage (Backblaze B2 or Cloudflare R2, both far below S3 for this profile). Working set as 2,048 px previews plus extracted metadata, ~600 GB, on local disk.
- **Content-addressed by sha256**, so a duplicate contribution costs nothing and re-download is verifiable.
- **The GOLDEN corpus is the exception:** 1,200 images must be reachable by CI on every run. Store them as compressed DNG in a dedicated bucket with a pinned checksum manifest, and cache them on runners. Budget ~40 GB.
- **A dataset version is a hash of the manifest**, recorded in every model card. "Trained on `corpus@a91f3c2`" must be reproducible three years from now, or the model is not reproducible either.

## 8. Tests and gates

- ☐ **Leakage test:** no `wedding_id` and no `sha256` appears in more than one corpus. Runs in `model-gate`; failure blocks the model.
- ☐ **Near-duplicate test:** perceptual hash distance across corpus boundaries above a threshold. Catches the same photo exported twice at different sizes.
- ☐ **Manifest completeness:** every file on disk has a row; every row has a licence reference that resolves to an existing document.
- ☐ **Withdrawal test:** marking a wedding `withdrawn` removes it from the next training run's file list, verified by asserting the list changed.
- ☐ **Coverage report:** generated per run; under-target slices listed explicitly, not silently rounded away.

- ☐ **Licence audit:** a quarterly job that re-reads every `licence_ref` and flags expired or unsigned agreements.
- ☐ **DEMO strictness:** every image with `corpus = 'DEMO'` has `model_release = 1`; a shipped build containing a demo image without a release fails the release gate.

## 9. Acceptance criteria

- ☐ `corpus/manifest.sqlite` exists, matches the frozen schema, and is under contract freeze.
- ☐ Every EVAL and GOLDEN wedding has a `corpus_split_lock` row.
- ☐ The coverage table meets every minimum in section 5, or each shortfall has a named owner and a date.
- ☐ At least 15 signed partner agreements are on file before the first model ships.
- ☐ Inter-annotator agreement is published per slice, and no model gate demands accuracy above human agreement on that slice.
- ☐ A dataset version hash appears in every model card in `models/*/CARD.md`.
- ☐ The leakage test has been demonstrated to fail when leakage is deliberately introduced.

## 10. Claude Code prompt

Act as `DATA` (Data Engineering Lead) with `SEC` (Security & Privacy) reviewing. Implement `23-DATASET-AND-CORPUS-SPEC.md`: create the manifest schema exactly as frozen in section 4, the ingestion CLI that refuses any image without a licence reference, the wedding-level splitter that writes `corpus_split_lock`, the leakage and near-duplicate tests, and the coverage report generator.

Constraints: the splitter must split by `wedding_id` and hold out whole photographers — never by image. Ingestion must be idempotent and content-addressed. No image may be moved between corpora by code; that requires an ADR.

Then prove the leakage test works by deliberately inserting a leaked wedding and showing the test fail, then removing it and showing it pass. Paste both

| outputs.